Main results

Cesàro Summation

A series $\sum a_n$ with partial sums $S_N = \sum_{k=0}^{N} a_k$ is *Cesàro summable* (C,1) to L if

$$\sigma_N = \frac{1}{N+1} \sum_{n=0}^{N} S_n \;\longrightarrow\; L.$$

Every convergent series is Cesàro summable to the same limit, but not conversely.

Positive Summation Kernels

A family of functions $(K_n)_{n \geq 1}$ on $\mathbb{R}$ (or on $\mathbb{T}$, the circle) is called a *positive summation kernel* if the following hold:

(i) $K_n(x) \geq 0$ for all $x \in \mathbb{R}$ and each n;

(ii) $\displaystyle\int_{-\infty}^{\infty} K_n(x)\,dx = 1$ for all n;

(iii) for every $\delta > 0$,

$$\int_{|x| \geq \delta} K_n(x)\,dx \;\longrightarrow\; 0 \quad \text{as } n \to \infty.$$

Equivalently, (K_n) is an *approximate identity*: the mass of K_n concentrates near $x = 0$ as $n \to \infty$. For such kernels, the convolution $(f * K_n)(x) = \int f(x-t)K_n(t)\,dt$ satisfies

$$(f * K_n)(x) \to f(x)$$

for all x where f is continuous.

Riemann-Lebesgue Lemma

Theorem 2.2 (Riemann–Lebesgue lemma) Let f be absolutely integrable in the Riemann sense on a finite or infinite interval I. Then

$$\lim_{\lambda \to \infty} \int_I f(u) \sin(\lambda u)\,du = 0.$$

(A corresponding result holds for $\cos(\lambda u)$ or $e^{i\lambda u}$.)

Corollary for Fourier Coefficients

Corollary. Let $f \in L^1(\mathbb{T})$ (i.e., $\int_{-\pi}^{\pi} |f(x)|dx < \infty$). Its complex Fourier coefficients are:

$$\hat{f}(n) = c_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-inx}\,dx, \qquad n \in \mathbb{Z}.$$

The Fourier coefficients of any integrable function vanish at infinity:

$$\hat{f}(n) \longrightarrow 0 \quad \text{as } |n| \to \infty.$$

Proof Sketch of Corollary: The Fourier coefficient $\hat{f}(n)$ can be expressed as:

$$\hat{f}(n) = \frac{1}{2\pi}\left(\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx + i\int_{-\pi}^{\pi} f(x)\sin(-nx)\,dx\right)$$

Since $\cos(nx)$ and $\sin(nx)$ are bounded functions, the absolute integrability of f is preserved when multiplied by these functions. As $|n| \to \infty$, both the real and imaginary parts of the integral tend to zero by applying the Riemann–Lebesgue lemma (and its cosine variant) with $\lambda = n$. Therefore, $\hat{f}(n) \to 0$ as $|n| \to \infty$.

Convolution (on $\mathbb{R}_+$) and the Convolution Theorem

Assume $f, g : \mathbb{R}_+ \to \mathbb{C}$ are causal (zero for $t < 0$), piecewise continuous, and of exponential order so that Laplace transforms exist.

Definition (Convolution)

$$(f * g)(t) \;:=\; \int_0^t f(t-\tau)\,g(\tau)\,d\tau, \qquad t \geq 0.$$

Algebraic Laws

For all admissible f, g, h and scalars α, β:

$$\begin{aligned} \text{Commutative:}\;& f * g = g * f, \\ \text{Associative:}\;& (f * g) * h = f * (g * h), \\ \text{Distributive:}\;& f * (\alpha g + \beta h) = \alpha(f * g) + \beta(f * h). \end{aligned}$$

Convolution Theorem (Laplace)

Let $F(s) = \mathcal{L}\{f\}(s)$ and $G(s) = \mathcal{L}\{g\}(s)$ on a common half-plane $\Re s > \sigma_0$. Then

$$\mathcal{L}\{f * g\}(s) = F(s)\,G(s)$$

(Convergence holds on any half-plane where both F and G exist.)

The Laplace Transform

Let $H(t)$ be the Heaviside step function, $H(t) = 0$ for $t < 0$ and $H(t) = 1$ for $t \geq 0$. We will always work with *causal* functions, i.e. we take f to be defined as

$$f(t) \ := \ f(t)\, H(t),$$

so that $f(t) = 0$ for all $t < 0$.

For a complex parameter s with $\Re s$ large enough so that the integral converges (the *region of convergence*), the *Laplace transform* of f is

$$\mathcal{L}\{f\}(s) \ = \ \int_0^\infty f(t)\, e^{-st}\, dt.$$

We also write $\tilde{f}(s)$ or $F(s)$ for $\mathcal{L}\{f\}(s)$.

Functions of Exponential Order

Let $k > 0$. A function $f : [0, \infty) \to \mathbb{C}$ is said to be of *exponential order k* if:

(i) f is continuous on every finite subinterval of $[0, \infty)$, except possibly for finitely many jump discontinuities;

(ii) there exists a constant $M > 0$ such that

$$|f(t)| \leq M e^{kt}, \qquad t \geq 0.$$

The class of all such functions is denoted by $\mathcal{E}_k$. If $f \in \mathcal{E}_k$ for some k, we say that f is of *exponential type* and write $f \in \mathcal{E} = \bigcup_{k>0} \mathcal{E}_k$.

Theorem. If $f \in \mathcal{E}_k$, then the Laplace transform

$$\tilde{f}(s) = \int_0^\infty f(t) e^{-st}\, dt$$

exists for all $s > k$.

Common Laplace Transforms

(Here $a, b \in \mathbb{R}$, $n \in \mathbb{N}$, and the stated condition on s ensures convergence.)

Function $f(t)$	$\mathcal{L}\{f\}(s)$	Condition on s
1	$\frac{1}{s}$	$\Re s > 0$
t^n	$\frac{n!}{s^{n+1}}$	$\Re s > 0$
e^{at}	$\frac{1}{s-a}$	$\Re s > a$
$\sin(bt)$	$\frac{b}{s^2+b^2}$	$\Re s > 0$
$\cos(bt)$	$\frac{s}{s^2+b^2}$	$\Re s > 0$
$\sinh(bt)$	$\frac{b}{s^2-b^2}$	$\Re s > \lvert b\rvert$
$\cosh(bt)$	$\frac{s}{s^2-b^2}$	$\Re s > \lvert b\rvert$
$f(t-a)\,H(t-a)$	$e^{-as}F(s)$	$a > 0$ (second shifting)
$e^{at}f(t)$	$F(s-a)$	$\Re s > a+\sigma_0$, if F converges for $\Re s > \sigma_0$

Operational Rules for the Laplace Transform

Let $\mathcal{L}\{f(t)\}(s) = \tilde{f}(s)$, and assume all transforms exist for sufficiently large $\Re s$.

(i) **Linearity:**

$$\mathcal{L}\{\alpha f(t) + \beta g(t)\}(s) = \alpha \tilde{f}(s) + \beta \tilde{g}(s), \qquad \alpha, \beta \in \mathbb{C}.$$

(ii) **Exponential (Damping) Rule:**

$$\mathcal{L}\{e^{at}f(t)\}(s) = \tilde{f}(s-a).$$

(iii) **Time Shift (Delay) Rule:** If $f(t) = 0$ for $t < 0$ and $a > 0$, then

$$\mathcal{L}\{f(t-a)\}(s) = e^{-as}\,\tilde{f}(s).$$

(iv) **Scaling Rule:** For $a > 0$,

$$\mathcal{L}\{f(at)\}(s) = \frac{1}{a}\,\tilde{f}\left(\frac{s}{a}\right).$$

(v) **Multiplication by t (and by t^n):**

$$\mathcal{L}\{t\,f(t)\}(s) \;=\; -\frac{d}{ds}\tilde{f}(s), \qquad \mathcal{L}\{t^n f(t)\}(s) \;=\; (-1)^n \frac{d^n}{ds^n}\tilde{f}(s) \quad (n \in \mathbb{N}).$$

(vi) **Derivatives in t:**

$$\mathcal{L}\{f'(t)\}(s) = s\,\tilde{f}(s) - f(0^+), \qquad \mathcal{L}\{f''(t)\}(s) = s^2\tilde{f}(s) - sf(0^+) - f'(0^+),$$

and in general

$$\mathcal{L}\{f^{(n)}(t)\}(s) = s^n\tilde{f}(s) - \sum_{k=0}^{n-1} s^{\,n-1-k} f^{(k)}(0^+).$$

(vii) **Integral in t:**

$$\mathcal{L}\left\{\int_0^t f(u)\,du\right\}(s) = \frac{\tilde{f}(s)}{s}.$$

Fourier Series Coefficients

The representation of a 2π-periodic function f as a Fourier series depends on whether the complex exponential basis or the real trigonometric basis is used.

1. Complex Fourier Series

For a function $f \in L^1(\mathbb{T})$, the series representation is:

$$f(x) \sim \sum_{n \in \mathbb{Z}} c_n e^{inx}$$

The complex Fourier coefficients $c_n = \hat{f}(n)$ are computed as:

$$c_n = \hat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx}\, dx, \qquad n \in \mathbb{Z}.$$

2. Real Fourier Series (Trigonometric)

For a real-valued function f, the series representation is:

$$f(x) \sim a_0 + \sum_{n=1}^{\infty} \left(a_n \cos(nx) + b_n \sin(nx)\right)$$

The real Fourier coefficients are computed as:

- a_0 (Constant component):
$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\, dx$$
- a_n (Cosine coefficients, $n \geq 1$):
$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx)\, dx, \qquad n \in \mathbb{N}$$
- b_n (Sine coefficients, $n \geq 1$):
$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx)\, dx, \qquad n \in \mathbb{N}$$

Note: The relationship between the coefficients is $c_0 = a_0$, and for $n \geq 1$, $c_n = \frac{1}{2}(a_n - ib_n)$ and $c_{-n} = \frac{1}{2}(a_n + ib_n)$.

Fourier Inversion Theorem (Periodic Case)

Let $f : \mathbb{R} \to \mathbb{C}$ be a 2π-periodic function such that f is piecewise continuous and of bounded variation on $[-\pi, \pi]$. Its (complex) Fourier coefficients are

$$\hat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-int}\, dt, \qquad n \in \mathbb{Z}.$$

Then the Fourier series

$$S_N(f)(t) = \sum_{|n|\leq N} \hat{f}(n)e^{int}$$

converges, for each t, to

$$\lim_{N\to\infty} S_N(f)(t) = \frac{f(t^-)+f(t^+)}{2}.$$

That is, the series converges to the midpoint of the left and right limits of f at each point. In particular, if f is continuous at t, then

$$\sum_{n\in\mathbb{Z}} \hat{f}(n)e^{int} = f(t).$$

Consequences

- If $\sum\limits_{n\in\mathbb{Z}} |\hat{f}(n)| < \infty$, the Fourier series converges *uniformly* to a *continuous* 2π-periodic function.
- If f has a jump discontinuity at t_0, the Fourier series converges to $\frac{1}{2}(f(t_0^-)+f(t_0^+))$, not to either one-sided limit.
- Evaluating at $t=0$ gives $\sum_{n\in\mathbb{Z}} \hat{f}(n) = \frac{f(0^-)+f(0^+)}{2}$,

Fourier Transform (on $\mathbb{R}$)

Definition and Convention

For $f \in L^1(\mathbb{R})$, the Fourier Transform $\hat{f}$ is defined as:

$$\hat{f}(\omega) = \mathcal{F}\{f\}(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t}dt, \qquad \omega \in \mathbb{R}.$$

Real and Imaginary Parts

For a real-valued function $f(t)$, the Fourier Transform $\hat{f}(\omega)$ is generally complex and can be split into real and imaginary parts:

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(t)(\cos(-\omega t)+i\sin(-\omega t))dt = \int_{-\infty}^{\infty} f(t)\cos(\omega t)dt - i\int_{-\infty}^{\infty} f(t)\sin(\omega t)dt$$

Thus, we define:

$$\Re\{\hat{f}(\omega)\} = \int_{-\infty}^{\infty} f(t)\cos(\omega t)dt$$

$$\Im\{\hat{f}(\omega)\} = -\int_{-\infty}^{\infty} f(t)\sin(\omega t)dt$$

Symmetry Simplifications (for Real-Valued f)

The symmetry of $f(t)$ simplifies the computation of $\hat{f}(\omega)$.

- ($f(t) = f(-t)$): The product $f(t)\sin(\omega t)$ is odd, so $\int_{-\infty}^{\infty} f(t)\sin(\omega t)dt = 0$. The product $f(t)\cos(\omega t)$ is even, so $\int_{-\infty}^{\infty} f(t)\cos(\omega t)dt = 2\int_0^{\infty} f(t)\cos(\omega t)dt$.

$$\hat{f}(\omega) = 2\int_0^{\infty} f(t)\cos(\omega t)dt \quad (\hat{f} \text{ is purely real and even})$$

- ($f(t) = -f(-t)$): The product $f(t)\cos(\omega t)$ is odd, so $\int_{-\infty}^{\infty} f(t)\cos(\omega t)dt = 0$. The product $f(t)\sin(\omega t)$ is even, so $\int_{-\infty}^{\infty} f(t)\sin(\omega t)dt = 2\int_0^{\infty} f(t)\sin(\omega t)dt$.

$$\hat{f}(\omega) = -i\left(2\int_0^{\infty} f(t)\sin(\omega t)dt\right) \quad (\hat{f} \text{ is purely imaginary and odd})$$

Key $L^1(\mathbb{R})$ Properties

Property	Statement
Riemann–Lebesgue	If $f \in L^1(\mathbb{R})$, then $\lim_{\lvert\omega\rvert\to\infty} \hat{f}(\omega) = 0$.
Continuity	If $f \in L^1(\mathbb{R})$, then $\hat{f}(\omega)$ must be a **continuous** function on $\mathbb{R}$.
Inversion	If $f, \hat{f} \in L^1(\mathbb{R})$, then $f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{f}(\omega)e^{i\omega t}d\omega$.

Operational Rules

Time Domain	Frequency Domain
$\mathcal{F}\{f'(t)\}$	$i\omega\hat{f}(\omega)$
$\mathcal{F}\{tf(t)\}$	$i\frac{d}{d\omega}\hat{f}(\omega)$
$\mathcal{F}\{f(t-a)\}$	$e^{-i\omega a}\hat{f}(\omega)$
$\mathcal{F}\{f * g\}$	$\hat{f}(\omega)\hat{g}(\omega)$ (Convolution Theorem)

Plancherel's Theorem (Parseval's Identity for Fourier Transforms)

Plancherel's Theorem relates the L^2 norm of a function in the time domain $L^2(\mathbb{R})$ to the L^2 norm of its Fourier Transform in the frequency domain $L^2(\mathbb{R})$. This identity is essential for evaluating integrals using Fourier transforms.

1. Squared L^2-Norm

For a square-integrable function $f \in L^2(\mathbb{R})$, the L^2-norm is preserved under the Fourier Transform:

$$\int_{-\infty}^{\infty} |f(t)|^2 dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |\hat{f}(\xi)|^2 d\xi$$

Note: The factor $\frac{1}{2\pi}$ depends on the convention used for the Fourier Transform.

2. Polarized Plancherel Formula

For $f, g \in L^2(\mathbb{R})$, the inner product is preserved:

$$\int_{-\infty}^{\infty} f(t)\overline{g(t)}dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{f}(\xi)\overline{\hat{g}(\xi)}d\xi$$

The integrals in Problem 6(b) often rely on applying these formulas, usually by identifying the integrand with either the square of a transform $|\hat{f}(\xi)|^2$ or a product of transforms $\hat{f}(\xi)\overline{\hat{g}(\xi)}$.

Canonical PDEs

Laplace Equation in the Unit Disk (Dirichlet Problem)

The Dirichlet problem for Laplace's equation ($\Delta u = 0$) on the unit disk $\mathbf{D} = \{r < 1\}$ is defined by:

$$\Delta u = 0 \text{ in } \{r < 1\}, \qquad u(1, \theta) = g(\theta).$$

The Poisson Kernel

The Poisson Kernel, $P_r(s)$ for the unit disk is defined by its Fourier series and its closed form:

$$P_r(s) := \frac{1}{2\pi}\sum_{n\in\mathbb{Z}} r^{|n|}e^{ins} = \frac{1}{2\pi}\frac{1 - r^2}{1 + r^2 - 2r\cos s}.$$

The General Solution

The unique solution $u(r, \theta)$ is given by the Poisson Integral Formula (a convolution of the boundary data $g(\phi)$ with the Poisson Kernel):

$$u(r, \theta) = \int_{-\pi}^{\pi} P_r(\theta - \phi)\, g(\phi)\, d\phi.$$

Alternatively, the solution is expressed as a Fourier series by multiplying the boundary function's coefficients with r^n:

$$u(r, \theta) = \frac{a_0}{2} + \sum_{n=1}^{\infty} r^n\big(a_n \cos n\theta + b_n \sin n\theta\big),$$

where a_n and b_n are the Fourier coefficients of $g(\theta)$.

Properties of the Poisson Kernel

The Poisson Kernel $P_r(s)$ has the following key properties for $r < 1$:

1. $P_r(s) = P_r(-s) \geq 0$.
2. $\int_{\mathbf{T}} P_r(s)\, ds = 1$.
3. If $\delta > 0$, then $\lim_{r\nearrow 1} \int_{\delta}^{\pi} P_r(s)\, ds = 0$.

Heat Equation on the Infinite Rod

The problem seeks the temperature $u(x,t)$ in an infinite rod given an initial temperature distribution $f(x)$. This is the initial value problem (IVP):

$$u_{xx} = u_t, \quad t > 0,\ x \in \mathbb{R} \qquad \text{with} \qquad u(x,0) = f(x).$$

The problem is solved using the Fourier Transform in the spatial variable x, which converts the PDE into a first-order ODE in time in the transform domain.

The General Solution (Convolution)

The unique solution $u(x,t)$ is given by the convolution of the initial data $f(x)$ with the **Heat Kernel** $E(x,t)$:

$$u(x,t) = (E * f)(x) = \frac{1}{\sqrt{4\pi t}} \int_{-\infty}^{\infty} e^{-y^2/(4t)} f(x-y)\, dy, \quad t > 0.$$

The Heat Kernel $E(x,t)$

The Heat Kernel, also known as the fundamental solution, is the function whose Fourier Transform is $e^{-\omega^2 t}$. It is defined as:

$$E(x,t) = \frac{1}{\sqrt{4\pi t}} \exp\left(-\frac{x^2}{4t}\right).$$

This kernel is a **positive summation kernel** (or approximate identity). This means that for fixed $t > 0$, $E(x,t)$ is positive, and as $t \to 0$: This ensure that $u(x,t)$ approaches the initial condition $f(x)$ as $t \to 0$.